

Global Filters

Filter by Market Location

- Pacific Asia xLATAM x
- USCA xEurope x
- Africa x

Filter by Order Region

- South Asia x
- Central America x
- East of USA x
- Southern Europe x
- Western Europe x

Filter by Segment/Tier

- Medium-Value Tier x
- Loss-Making Tier x
- High-Value Tier x

Filter by Category Name

- Cardio Equipment x
- Shop By Sport x
- Water Sports x
- Women's Apparel x
- Camping & Hiking x

Filter by Product Name

- Rock music x
- Adult dog supplies x
- Smart watch xToys x
- Fighting video games x
- Summer dresses x

Filter by Order Discount Rate Range

0.060.22

 APL Logistics Financial Performance Dashboard

 1. Financial Performance Overview

Total Revenue

Total Profit

Overall Profit Margin

\$22,482,297.22

\$2,335,911.77

10.39%

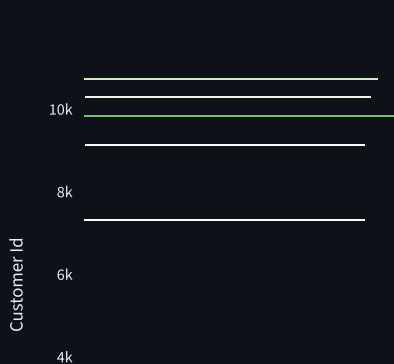
Regional Revenue vs Profit Insights

Revenue vs. Net Profit Concentration Across Order Regions



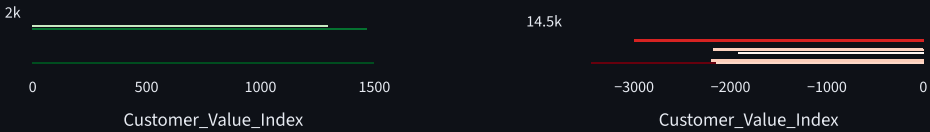
 2. Customer Contribution & Value Index

Top 10 High-Value Customers (By Profit)



Top 10 Loss-Making Customers





Value Tier Distribution Segment Analysis

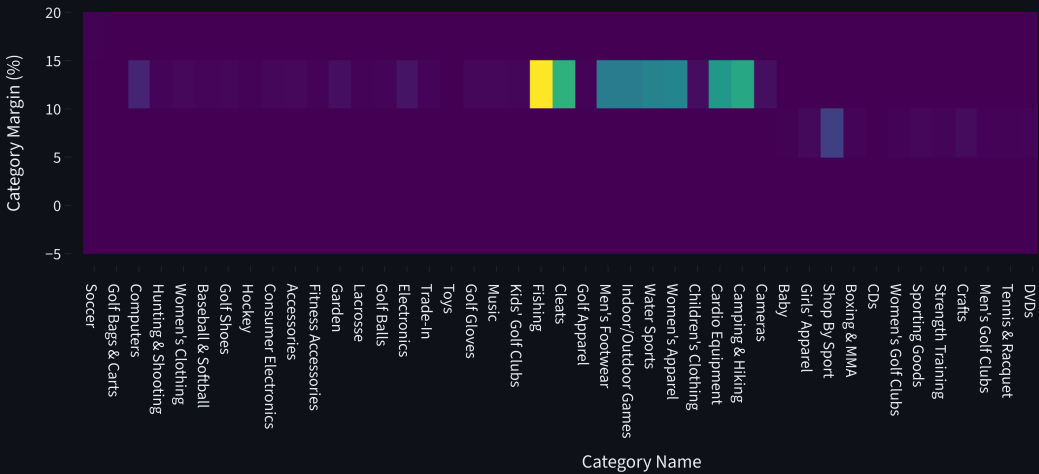
	Value Tier	Total_Sales	Total_Profit
0	High-Value Tier	\$439,866.75	\$150,997.32
1	Loss-Making Tier	\$4,196,473.63	\$-2,314,011.87
2	Medium-Value Tier	\$17,845,956.84	\$4,498,926.32



3. Product Portfolio & Category Diagnostics

Category Performance & Heatmap View

Category Heatmap: Margin vs Revenue Scaling Size



Strategic Warning: High Revenue but Low Margin Products

	Product Name	Revenue	Profit
21	Diamondback Women's Serene Classic Comfort Bi	\$2,517,732.14	\$255,35
61	Nike Men's Free 5.0+ Running Shoe	\$2,236,276.35	\$227,79
102	Under Armour Girls' Toddler Spine Surge Runni	\$773,846.49	\$72,22
110	Web Camera	\$163,186.44	\$16,38
11	Children's heaters	\$141,411.60	\$14,43
74	Porcelain crafts	\$136,136.60	\$11,30
78	Smart watch	\$72,105.00	\$6,11
26	First aid kit	\$64,468.80	\$4,50
17	DVDs	\$48,656.48	\$3,81
57	Nike Men's Comfort 2 Slide	\$27,848.81	\$2,25



4. Discount Impact & Strategic Scenarios

Margin Degradation Analysis

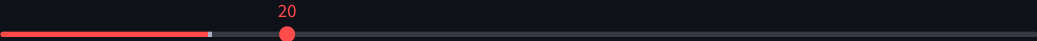
	Has_Discount	Revenue	Profit
0	☒	\$22,482,297.22	\$2,335,911



'What-If' Simulation Model

Simulate adjustments to current discounting logic to see changes to bottom line profitability.

Reduce existing discount rates across orders by (%):



Original Profit

\$2,335,911.77

Simulated Net Profit

\$2,918,462.12

Simulated Margin

12.98%

↑ \$582,550.35



Project Retrospective: Challenges, Tasks & Technical Summary



Project Context & Executed Tasks

The goal of this project was to establish an end-to-end analytics and forecasting pipeline for **APL Logistics** to evaluate revenue leaks, target logistical inefficiencies, and study how current promotional structures impact real profitability.

The core operations carried out during this project include:

- Financial Validation & Cleanup:** Auditing transactional metrics to filter out statistical anomalies and business-crippling structural inconsistencies.
- Customer Segmentation Tiers:** Isolating top-tier brand champions from high-volume, net loss-making commercial accounts using a dynamic *Customer Value Index*.

3. **Product & Category Diagnostic Mapping:** Auditing the product line to flag high-volume items suffering from thin margins due to overhead costs.
 4. **Predictive Risk Assessment:** Building standalone Machine Learning pipelines in Google Colab to gauge delivery and delay factors before they impact client relationships.
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Technical Challenges & Engineering Solutions

- **The Special Characters Encoding Problem:**
 - *Challenge:* The raw data contained international geographic labels, cities, and specific accent marks that caused immediate standard `utf-8` decoding failures during initialization.
 - *Solution:* Resolved by implementing a text-fallback mechanism using byte-resilient `latin1` encoding profiles during the Pandas parsing sequence.
 - **Target Leakage Risks in Machine Learning Modeling:**
 - *Challenge:* When engineering predictors for shipping delay risk, columns like actual shipment times and downstream delivery status flags created target leakage, synthetically inflating model validation metrics to an unrealistic 100% accuracy.
 - *Solution:* Identified and dropped leaky post-transaction columns during Phase 1 preprocessing to ensure the model makes predictions using true upstream supply chain information.
 - **Streamlit Runtime Lag & Refresh Delays:**
 - *Challenge:* Running multi-layered data cleaning logic, categorical matrix encodings, and string operations in real-time within the Streamlit code block degraded UI performance on interactive steps.
 - *Solution:* Structured a decoupled execution system where intensive financial pruning and feature narrowing are pre-computed in Google Colab, feeding a highly optimized, streamlined `.csv` payload into the visualization app.
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Technology Stack & Tools Used

- **Data Engineering & Preparation:** `Python`, `Pandas`, and `NumPy` for operational filters, categorical encoding transformation loops, and numerical cleaning matrix operations.
- **Predictive Frameworks:** `Scikit-Learn` utilizing balanced `RandomForestClassifier` configurations and robust scaling modules to address variable variance.
- **Interactive Visualization Suite:** `Streamlit Web Server Engine` combined with `Plotly Express` and `Plotly Graph Objects` to provide rich, interactive visualizations without performance lag.